# Sentra — Cron Job Monitoring Demo

Step-by-step commands to create a scheduled job, watch Sentra monitor it, simulate failures, and delete it — with the expected output of every step.

## How it works (say this first)

A scheduled (cron) job cannot be observed from outside: nothing on the network tells you whether last night's backup ran. So the job reports to Sentra itself. Each time it finishes successfully, it calls a secret **ping URL**. Sentra checks every 30 seconds whether the last ping is recent enough. If pings stop, or the job reports a failure, the check turns **DOWN**, an **incident** opens, and alerts are sent. This is called a **dead man's switch**, and it is the **push** model: the monitored job calls Sentra, Sentra never logs into the job's machine.

```
Windows Task Scheduler (every minute)
   └─ runs demo-cron-job.ps1  ── zips the docs folder (a fake "backup")
        ├─ start     ──►  /ping/<token>/start
        ├─ success   ──►  /ping/<token>          ──►  Sentra: "last ping just now" → HEALTHY
        └─ failure   ──►  /ping/<token>/fail     ──►  Sentra: DOWN + incident (immediately)
   no ping at all for (period + allowed delay)   ──►  Sentra: DOWN + incident
```

### What the demo shows

1. **Part A** — Start Sentra (database, backend, dashboard)
2. **Part B** — Create the job: Steps 1–6
3. **Part C** — Monitoring and failure demos: Steps 7–8
4. **Part D** — Delete the job: Step 9
5. **Part E** — Troubleshooting

## Part A — Start Sentra (do this before the teacher arrives)

Use three separate PowerShell windows. Leave them all open during the demo.

### A1. Start the database

Open **Docker Desktop** and wait until it says it is running. Then:

```
cd C:\Users\shubh\OneDrive\Documents\Sentra
docker compose up -d
```

**MEANING**

Starts PostgreSQL in a container on port 55432. All checks, runs and incidents are stored here.

**Expected:** `Container sentra-db Started` (or `Running` if it was already up).

### A2. Start the backend (window 1)

```
cd C:\Users\shubh\OneDrive\Documents\Sentra\backend
mvn spring-boot:run "-Dspring-boot.run.profiles=dev" `
    "-Dspring-boot.run.jvmArguments=-Xmx512m" `
    "-Dspring-boot.run.arguments=--server.port=8081"
```

The `` ` `` (backtick) at the end of a line means "the command continues on the next line".

**MEANING**

Runs the Spring Boot server on port 8081. It contains the scheduler that checks every job every 30 seconds.

**Expected:** after ~15 seconds the log ends with `Tomcat started on port 8081` and `Started SentraApplication`. Leave this window open.

If `mvn` is not recognised, run this first in the same window:

```
$machine = [Environment]::GetEnvironmentVariable("Path","Machine")
$user    = [Environment]::GetEnvironmentVariable("Path","User")
$env:Path = "$machine;$user"
```

### A3. Start the dashboard (window 2)

```
cd C:\Users\shubh\OneDrive\Documents\Sentra\frontend
npm run dev
```

**Expected:** `VITE ... ready` and a line `Local: http://localhost:5173/` (or 5174 if 5173 is busy). Open that address in the browser and log in.

### A4. Check the backend is alive (window 3 — use this window for every step below)

```
curl.exe http://localhost:8081/api/v1/ping
```

**Expected:** `{"status":"ok"}`

## Part B — Create the demo job

### Step 1. Create the check in Sentra (browser)

1. In the dashboard, click **+ Add check**.
2. Choose the ⏱ **Cron job** type at the top of the form.
3. Name: `Demo backup job`
4. Job runs every: **1 minute** · Allowed delay (minutes): **1**
5. Click **Add check**.

MEANING

Tells Sentra: "expect a ping from this job every minute; if none arrives within 1 minute plus 1 minute of slack, raise the alarm."

**Expected:** the check's page opens with a **Ping URL** such as `http://localhost:5173/ping/AbC123xyz...`. State is **PENDING**, "Last ping: none yet". Copy the **token** — the long text after `/ping/`.

*"Sentra generated a random, unguessable token. Possessing this URL is the job's only credential."*

### Step 2. Save the token in a variable

```
$token = "PASTE-YOUR-TOKEN-HERE"
```

**Expected:** no output.

### Step 3. Send one ping by hand

```
curl.exe http://localhost:8081/ping/$token
```

MEANING

You pretend to be the job and report "I ran". This proves the URL and token are correct.

**Expected:** `OK`. In the browser, "Last ping" shows **just now** and within ~30 s the state becomes **HEALTHY**.
`404` means the token was copied wrongly — redo Step 2.

### Step 4. Create the scheduled task (runs every minute)

```
$script = "C:\Users\shubh\OneDrive\Documents\Sentra\scripts\demo-cron-job.ps1"
$run  = "conhost.exe --headless powershell.exe -NoProfile -ExecutionPolicy Bypass"
$run += " -File $script -Token $token"
schtasks /Create /TN "SentraDemoJob2" /SC MINUTE /MO 1 /TR $run /F
```

MEANING

Windows Task Scheduler is Windows' version of cron. The task runs the demo script every minute, hidden (no window). The script zips the project's `docs` folder as a pretend backup, then pings Sentra.

**Expected:** `SUCCESS: The scheduled task "SentraDemoJob2" has successfully been created.`

### Step 5. Allow the task to run on battery

```
$settings = New-ScheduledTaskSettingsSet -AllowStartIfOnBatteries -DontStopIfGoingOnBatteries
Set-ScheduledTask -TaskName "SentraDemoJob2" -Settings $settings
```

MEANING

By default Windows skips scheduled tasks when a laptop is unplugged — the job would silently never run.

**Expected:** a short description of the task, no red error text.

*"This is exactly the kind of silent failure Sentra exists to catch: the job simply never ran, and nothing crashed."*

### Step 6. Run it once now and confirm

```
schtasks /Run /TN "SentraDemoJob2"
Get-Content "$env:TEMP\sentra-demo-backups\job.log" -Tail 3
```

**Expected:** `SUCCESS: Attempted to run the scheduled task "SentraDemoJob2".`
A new log line like `2026-10-28 18:30:05Z OK docs-20261028-183004.zip`.
In the dashboard the check is **HEALTHY**, and "Last ping" refreshes every minute on its own from now on.

## Part C — Monitoring and failure demos

### What to point at on the dashboard

| On screen | What it means |
|---|---|
| **PENDING** | Created, no ping yet — Sentra does not alert for a job that has not had a chance to run. |
| **HEALTHY** (green) | Last ping arrived on time and reported success. |
| **DOWN** (red) | Ping is late, or the job reported a failure. An incident is open. |
| Last ping | When the job last reported; "(reported a failure)" if it called `/fail`. |
| Run tape / Recent runs | Sentra's own check every 30 s — green = on time, red = late or failed. |
| Incidents | One row per outage: when it opened, how long it lasted, the reason. |

### Step 7. Demo: the job runs but fails

```
New-Item -ItemType File "$env:TEMP\sentra-demo-backups\FAIL"
schtasks /Run /TN "SentraDemoJob2"
```

MEANING

The `FAIL` file makes the script fail on purpose (like a full disk or database error). The script then calls `/ping/<token>/fail`.

**Expected:** the log shows `FAIL simulated failure ...`. Within seconds the check turns **DOWN** and an **Open** incident appears: "Job reported a failure".

**Fix it:**

```
Remove-Item "$env:TEMP\sentra-demo-backups\FAIL"
schtasks /Run /TN "SentraDemoJob2"
```

**Expected:** a new `OK` log line; within ~30 s the check is **HEALTHY** and the incident shows **Resolved** with its duration.

### Step 8. Demo: the job stops running completely

```
schtasks /Change /TN "SentraDemoJob2" /DISABLE
```

MEANING

Simulates a crashed server, a deleted crontab, or a job that silently stopped. No error is produced anywhere — the only signal is that the pings stop.

**Expected:** "Last ping" stops updating. After about **2–2.5 minutes** (1 min period + 1 min allowed delay + up to 30 s until Sentra's next check) the check turns **DOWN**: "No ping for ..., expected every 1 min".

*"You cannot detect the absence of an event by watching for an event. Sentra expects the ping, so silence itself becomes the alarm."*

**Fix it:**

```
schtasks /Change /TN "SentraDemoJob2" /ENABLE
schtasks /Run /TN "SentraDemoJob2"
```

**Expected:** within ~30 s the check is **HEALTHY** again and the incident is **Resolved**.

### Useful commands during the demo

```
# What the job did recently
Get-Content "$env:TEMP\sentra-demo-backups\job.log" -Tail 5

# Task status: last run time, last result, next run time
schtasks /Query /TN "SentraDemoJob2" /V /FO LIST

# The backup zips the job made
Get-ChildItem "$env:TEMP\sentra-demo-backups"
```

## Part D — Delete the job

### Step 9. Remove the scheduled task and the check

```
schtasks /Delete /TN "SentraDemoJob2" /F
```

**Expected:** `SUCCESS: The scheduled task "SentraDemoJob2" was successfully deleted.`

Then open the check in the dashboard and scroll to **Manage check**:

- ⏸ **Pause** stops monitoring and keeps the history (use ▶ **Resume** to restart it), or
- **Delete check** → **Yes, delete** removes the check with all its runs and incidents.

**Expected:** after deleting, you land back on the dashboard and the check is gone. Without this step the check would go **DOWN** after ~2 minutes, because the pings have stopped (which is the correct behaviour).

Optional — remove the demo backups and log:

```
Remove-Item "$env:TEMP\sentra-demo-backups" -Recurse -Force
```

## Part E — Troubleshooting

| Problem | Cause and fix |
|---|---|
| `mvn` not recognised | Terminal opened before Maven was installed. Run the PATH line in A2, or restart the terminal/IDE. |
| Backend: "Port 8081 was already in use" | A backend is already running. Use it, or close the other window first. |
| Backend can't connect to database | Docker Desktop not running or container stopped. Redo A1. |
| `curl` gives 404 | Wrong or incomplete token. Copy it again from the check page (Step 2). |
| `curl` gives 403 / "Whitelabel Error Page" | The URL is missing `/ping/<token>`. Only that path is public. |
| Check is DOWN but the task "should" be running | Laptop on battery and Step 5 skipped, or the task is disabled. Check with `schtasks /Query ...`; "Last Run Time 1999" means it never ran. |
| Two demo jobs fail together | The `FAIL` file is shared by every job that runs `demo-cron-job.ps1`. Delete it to recover all of them. |
| Task runs but no ping arrives | If the token starts with `-`, PowerShell may read it as an option. Create a new check to get a different token. |